

Homework 1

EE 474 — Introduction to Control Systems

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Due Wednesday, September 16, 2026

Problem 1.1b — Drebbel incubator — feedback loop

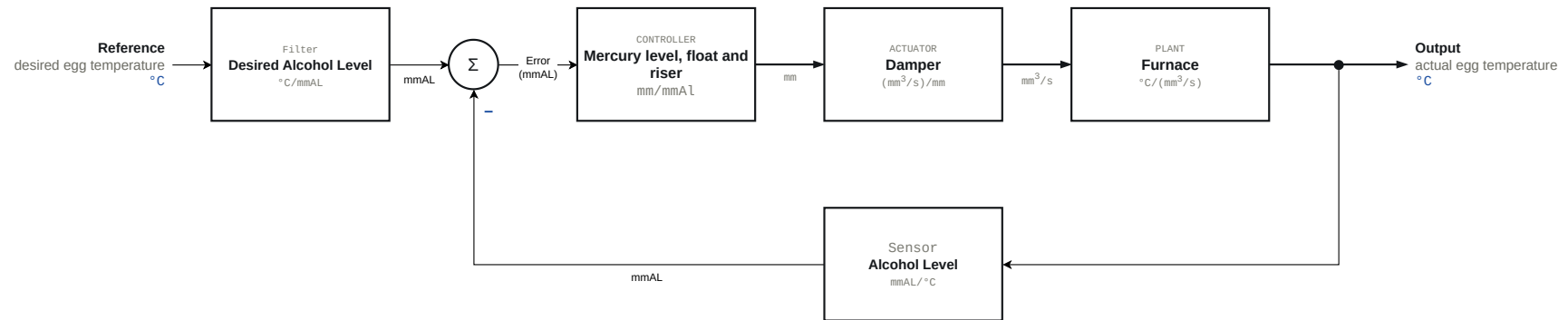


Figure 1: Block diagram of Drebbel's incubator drawn as a closed-loop control system.

Problem 2.A — Vector operations

Code: hw_01_A.m

```
1 % Initial Vectors
2 a = [ 1 2 4 2 1];
3 b = [4 3 -1 0 1];
4
5 disp("A.1")
6 a+b
7
8 disp("A.2")
9 a - b
10
11 disp("A.3")
12 min(a)
13
14 disp("A.4")
15 (a>b)
16
17 disp("A.5")
18 (a==b)
19
20 disp("A.6")
21 sin(a)
22
23 disp("A.7")
24 exp(b)
```

Output: output/hw_01_A.txt

```
A.1
ans =

    5    5    3    2    2

A.2
ans =

   -3   -1    5    2    0

A.3
ans = 1

A.4
ans =

    0    0    1    1    0

A.5
ans =

    0    0    0    0    1

A.6
ans =

    0.8415    0.9093   -0.7568    0.9093    0.8415
```

A.7

ans =

54.5982	20.0855	0.3679	1.0000	2.7183
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Problem 2.B — Matrix operations

Code: hw_01_B.m

```
1 a = [1 2 3; 3 2 3; 0 1 0];
2 b = [ 0 0 1; 0 1 1; 1 1 0];
3 c = [1 2 3];
4
5 disp("B.1")
6 a + b
7
8 disp("B.2")
9 d = inv(b)
10
11 disp("B.3")
12 b * d
13
14 disp("B.4")
15 sin(a)
16
17 disp("B.5")
18 a * b
19
20 disp("B.6")
21 a .* b
22
23 disp("B.7")
24 a'
25
26 disp("B.8")
27 a + a'
```

Output: output/hw_01_B.txt

```
B.1
ans =

     1     2     4
     3     3     4
     1     2     0

B.2
d =

     1    -1     1
    -1     1     0
     1     0     0

B.3
ans =

     1     0     0
     0     1     0
     0     0     1

B.4
ans =
```

0.8415	0.9093	0.1411
0.1411	0.9093	0.1411
0	0.8415	0

B.5

ans =

3	5	3
3	5	5
0	1	1

B.6

ans =

0	0	3
0	2	3
0	1	0

B.7

ans =

1	3	0
2	2	1
3	3	0

B.8

ans =

2	5	3
5	4	4
3	4	0